

Technology Stack

Project Title: Rising Waters

Introduction

The **Rising Waters** Flood Prediction System is developed using Machine Learning and web technologies. Each technology plays an important role in building, training, and deploying the application.

Programming Language

Python is used for data processing, model training, and application development.

Machine Learning

Scikit-learn and XGBoost are used to build and train the flood prediction model with high accuracy.

Frontend Technologies

HTML, CSS, and JavaScript are used to design a responsive and user-friendly web interface.

Backend Framework

Flask is used to connect the web interface with the Machine Learning model and process user requests.

Libraries

- Pandas – Data manipulation
- NumPy – Numerical computations
- Joblib – Model saving and loading

Development Tools

- Visual Studio Code
- Python IDLE / PyCharm
- GitHub (Version Control)

Conclusion

The selected technology stack provides a simple, efficient, and reliable platform for developing and deploying the **Rising Waters** Flood Prediction System.